

# Task 4

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## Scenario

The HR department wants reusable SQL objects to simplify reporting and improve data retrieval.

## Task

### Part 1 – User Defined Functions

1. Create a Scalar Function that returns the annual salary of an employee based on their monthly salary.
2. Create a Scalar Function that calculates the service years of an employee using their joining date.
3. Create a Table-Valued Function that returns all employees belonging to a specific department.

### Part 2 – SQL Queries

Using the functions created above, write SQL queries to:

1. Display Employee Name, Monthly Salary, and Annual Salary.
2. Display Employee Name and Years of Service.
3. Retrieve all employees from a department by calling the table-valued function.
4. Display employees whose annual salary is greater than 1,000,000.
5. Display the department with the highest average annual salary.

## Deliverables

- A single SQL script file (Day4\_SQL\_Task.sql) containing all functions and queries.
- Screenshots (or exported results) of the query outputs.

**Expected Completion Time:** 1 day.